

S.No. : 430

BCS 3602

No. of Printed Pages : 05

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 33226

Roll
No.

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B. Tech. Examination, 2024-25

(Even Semester)

DESIGN & ANALYSIS OF ALGORITHMS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION – A

1. Attempt all parts of the following : $8 \times 1 = 8$
- (a) Explain different characteristics of an algorithm.
 - (b) Explain different asymptotic notations.
 - (c) Explain difference between greedy and dynamic approach.
 - (d) Write three steps involved in divide and conquer approach for problem solving.
 - (e) Write down the best-case, worst case and average case-complexity of quicksort.

[P. T. O.

- (f) Explain flow networks.
- (g) Explain two properties of binomial heap.
- (h) Explain difference between BFS and DFS.

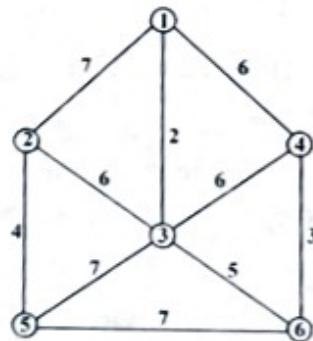
SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Sort the given array by using merge sort algorithm :

$\langle 5, 2, 4, 7, 1, 3, 2, 6 \rangle$

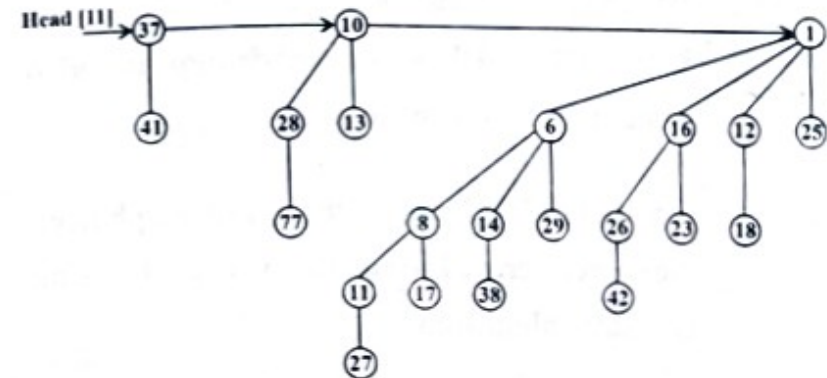
- (b) Explain Naive string matching algorithm with example.
- (c) Explain types of randomized algorithm with examples in detail.
- (d) Using Kruskal's algorithm find the minimum spanning tree of the following given graph :



SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Run the binomial-heap-extract-min procedure on the following binomial heap and give the resultant binomial heap :

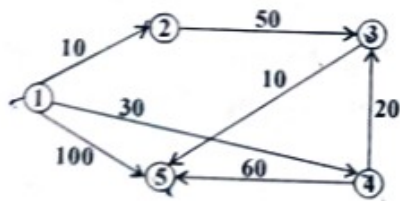


- (b) Explain in detail most common techniques used in amortized analysis.
- (c) Consider following four matrices. Find optimal parenthesization of matrix chain multiplication:

Matrix	A_1	A_2	A_3	A_4
Order	20×30	30×50	50×10	10×5

[P. T. O.]

4. (a) Explain quicksort algorithm and also write it's time and space complexity.
- (b) Explain the following :
- (i) P problem
 - (ii) NP problem
 - (ii) NP hard problem
- (c) Explain Flyod Warshall algorithm and also write it's time complexity.
5. (a) Find the shortest path in the below graph from the source vertex 1 to all other vertices by using Dijkstra's algorithm :



- (b) Explain depth first traversal method for graph with algorithm.
- (c) Explain approximation algorithm and apply the same on travelling salesperson problem.

6. (a) Solve following recurrences using Master theorem :
- (i) $T(n) = 2T\left(\frac{n}{2}\right) + n^3$
 - (ii) $T(n) = 16T\left(\frac{n}{4}\right) + n^2$
- (b) Explain KMP string matching algorithm.
- (c) Explain back tracking concept and draw the state space tree for 4 queens problem.
